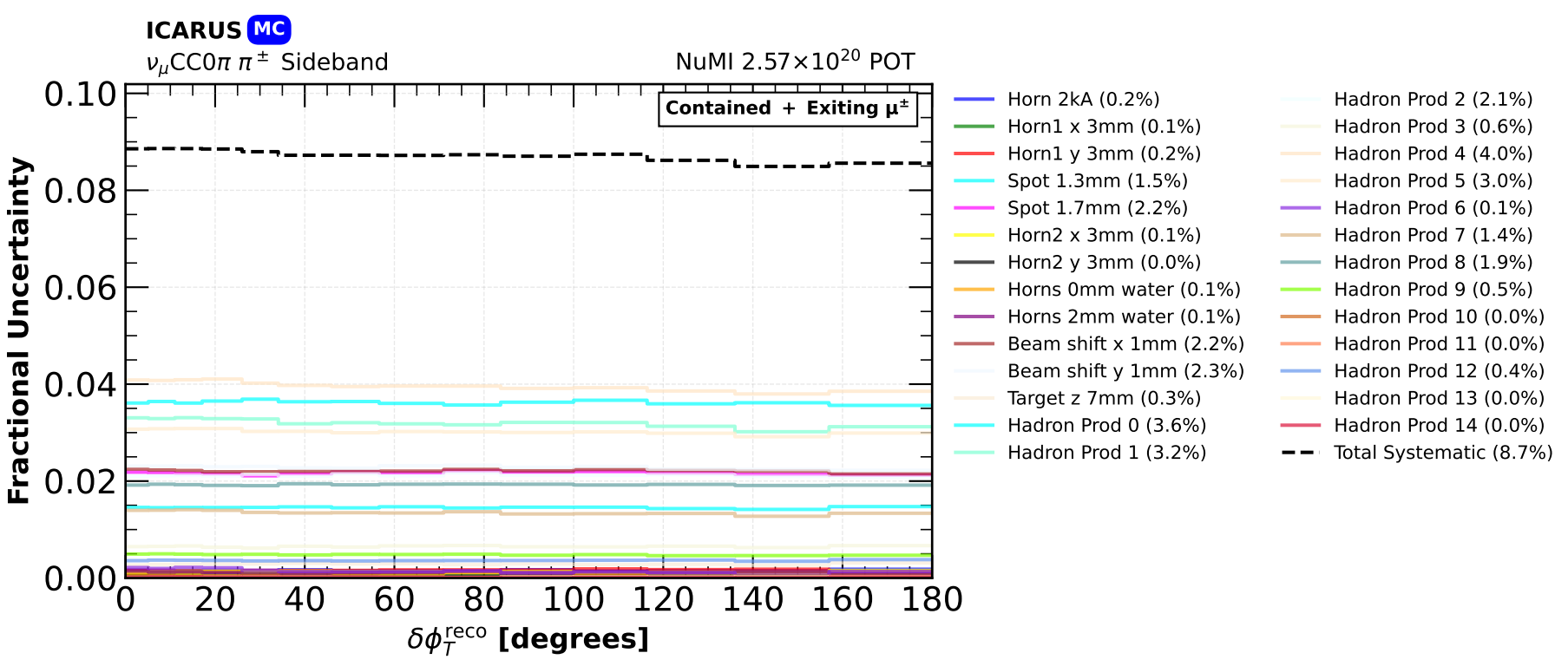
ICARUS MC
ν_μCC0π $\pi^\pm$ Sideband
NuMI 2.57×10^{20} POT
Contained + Exiting $\mu^\pm$
Fractional Uncertainty
0.10
0.08
0.06
0.04
0.02
0.00
0
20
40
60
80
100
120
140
160
180
$\delta\phi_T^{\text{reco}}$ [degrees]
Horn 2kA (0.2%)
Horn1 x 3mm (0.1%)
Horn1 y 3mm (0.2%)
Spot 1.3mm (1.5%)
Spot 1.7mm (2.2%)
Horn2 x 3mm (0.1%)
Horn2 y 3mm (0.0%)
Horns 0mm water (0.1%)
Horns 2mm water (0.1%)
Beam shift x 1mm (2.2%)
Beam shift y 1mm (2.3%)
Target z 7mm (0.3%)
Hadron Prod 0 (3.6%)
Hadron Prod 1 (3.2%)
Hadron Prod 2 (2.1%)
Hadron Prod 3 (0.6%)
Hadron Prod 4 (4.0%)
Hadron Prod 5 (3.0%)
Hadron Prod 6 (0.1%)
Hadron Prod 7 (1.4%)
Hadron Prod 8 (1.9%)
Hadron Prod 9 (0.5%)
Hadron Prod 10 (0.0%)
Hadron Prod 11 (0.0%)
Hadron Prod 12 (0.4%)
Hadron Prod 13 (0.0%)
Hadron Prod 14 (0.0%)
Total Systematic (8.7%)